

1.6 Classwork Solutions: Measurement review

Notation: $=$ joins values that are exactly equal; $\approx$ introduces a rounded value. Each conversion runs as one chain of unit factors and is evaluated once.

1. Significant digits.

(a) **3**

(c) **3**

(e) **4**

(b) **2**

(d) **4**

(f) **4**

(b) The leading zeros in 0.0062 only locate the decimal point. (c) The zero in 40.0 after the decimal point was measured, so it counts — and so does the zero between the 4 and it. (e) In 0.000 501 0 the four leading zeros do not count; the 5, the zero after it, the 1, and the final zero do. The digit grouping makes the leading zeros easy to count: there are four of them before the 5.

2. Round to three significant figures.

(a) $\sqrt{11} = 3.3166247 \dots \approx$ **3.32**

(b) 1609.344 m $\approx$ **1610 m**, better written **1.61×10^3 m**

The mile is defined as exactly 1609.344 m, so every digit is exact; the rounding is only because the question asks for three figures.

3. Scientific notation.

(a) 1,392,700 km $= 1.3927 \times 10^6$ km $\approx$ **1.39×10^6 km**

(b) 0.000 000 003 335 64 s $= 3.33564 \times 10^{-9}$ s $\approx$ **3.34×10^{-9} s**

Count the places the decimal point moves to land after the first nonzero digit: three groups of three gets you to the 3, so the exponent is -9 .

4. Ordinary decimal numbers.

(a) 7.8×10^{-6} m = **0.000 007 8 m**

(b) 1.496×10^8 km = **149,600,000 km**

5. Bolt's average speed.

$$v = \frac{100 \text{ m}}{9.58 \text{ s}} \times \frac{1 \text{ km}}{1000 \text{ m}} \times \frac{3600 \text{ s}}{1 \text{ h}} = 37.578 \dots \approx \textbf{37.6 km/h}$$

Three figures, set by the 9.58 s; the 100 m race distance is exact.

6. Reading the ruler. The bar ends between the 5.8 and 5.9 cm marks, about a third of the way along.

(a) **5.83 cm** — the 5.8 is certain and the final 3 is the estimated digit. *Accept 5.82 to 5.84.*

(b) **Three significant figures.**

$$(c) \ 5.83 \text{ cm} \times \frac{1 \text{ m}}{100 \text{ cm}} = \mathbf{0.0583 \text{ m}}$$

Still three significant figures — the leading zeros are placeholders.

7. Convert.

$$(a) \ 828 \text{ m} \times \frac{3.28 \text{ ft}}{1 \text{ m}} = 2715.84 \text{ ft} \approx \mathbf{2720 \text{ ft}}, \text{ or } \mathbf{2.72 \times 10^3 \text{ ft}}$$

$$(b) \ 20.5 \frac{\text{m}}{\text{s}} \times \frac{1 \text{ km}}{1000 \text{ m}} \times \frac{3600 \text{ s}}{1 \text{ h}} = \mathbf{73.8 \text{ km/h}}$$

Exactly 73.8, with no rounding needed: multiplying by 3.6 happens to land on three figures.

8. Fingernail growth.

$$3.5 \frac{\text{cm}}{\text{yr}} \times \frac{10 \text{ mm}}{1 \text{ cm}} \times \frac{1 \text{ yr}}{365.25 \text{ days}} \times \frac{7 \text{ days}}{1 \text{ week}} = 0.67077 \dots \approx \mathbf{0.67 \text{ mm per week}}$$

Two significant figures, because 3.5 cm has two. Reporting 0.671 mm would claim a precision the “about 3.5” does not have.